Supplementary Material: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

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TABLE S1
SUMMARY OF MATHEMATICAL NOTATIONS, TENSOR DIMENSIONS, AND OPERATORS.

Symbol	Mathematical Definition	Dimension / Type
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Continuous-time dynamic transaction graph	Graph object
$\mathbf{X} \in \mathbb{R}^{\lvert\mathcal{V}\rvert \times d_v}$	Node input attribute matrix	Real matrix
$\mathbf{E} \in \mathbb{R}^{\lvert\mathcal{E}\rvert \times d_e}$	Transaction edge feature matrix	Real matrix
Δt_{uv}	Continuous time delta ($t_v - t_u \geq 0$)	Non-negative scalar
$\Phi_{\text{time}}(\Delta t)$	Harmonic continuous temporal positional embedding	$\mathbb{R}^{d_t}$ vector
$\mathbf{g}_{ij} \in (0, 1)$	Context-aware adaptive edge trust gate	Scalar gate
$\Phi_{\text{flow}}(u, t)$	Mass flow conservation ratio	Scalar $\in [0, \infty)$
$\lambda_u(t)$	Hawkes process self-exciting transaction intensity	Non-negative scalar
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$	Temporally restricted forward and reverse PageRank taint vectors	$\mathbb{R}^{\lvert\mathcal{V}\rvert}$ vectors
$\mathbf{z}_{\text{inv}}(u, t)$	12-dimensional canonical representation	$\mathbb{R}^{12}$ vector
$\mathbf{h}_u^{(l)}$	Hidden state representation of node u at layer l	$\mathbb{R}^{d}$ vector
$\mathcal{C}_k$	Latent typology cluster of illicit transactions	Subset $\mathcal{C}_k \subset \mathcal{V}_{\text{illicit}}$
$\alpha_u \in [0, 1]$	Evidence-adaptive graph-tabular fusion weight	Scalar weight
$\mathbf{h}_u^{*}$	Unified anomaly representation vector	$\mathbb{R}^{d^*}$ vector
τ^*	Cost-sensitive Bayes risk decision threshold	Optimal scalar $\in (0, 1)$
$\hat{q}^{(y)}$	Class-conditional conformal quantile ($y \in \{0, 1\}$)	Calibration scalar
$\Gamma(X)$	Finite-sample conformal prediction set	Discrete subset $\subseteq \{0, 1\}$
α	Target significance error level ($1 - \alpha$ coverage)	Scalar $\in (0, 1)$

APPENDIX A
SUMMARY OF MATHEMATICAL NOTATIONS AND OPERATORS

Table S1 provides the comprehensive reference glossary of mathematical symbols, tensor dimensions, and operator definitions employed throughout C-STGB.

APPENDIX B
COMPLIANCE RULE VERIFIER AST GRAMMAR & SCHEMA SPECIFICATION

To guarantee regulatory determinism and enforce zero entity hallucinations during assisted Suspicious Activity Report (SAR) narrative draft generation (Section VI of the main paper), the multi-agent forensic swarm employs a formal Abstract Syntax Tree (AST) grammar and runtime Pydantic schema validator.

A. Extended Backus-Naur Form (EBNF) Grammar

All natural-language and structured FinCEN Form 111 XML narratives generated by the SAR Drafter Agent must strictly conform to the following formal grammar $\mathcal{G}_{\text{SAR}}$:

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```
SAR_Document   ::= Header
                   SubjectSection
                   ActivitySection
                   EvidenceSection
                   AuditTrailer
Header         ::= "<SAR_Form111"
                   " Version='2.0'>"
                   FilingInst
                   FilingTimestamp
                   CaseID
                   "</SAR_Form111>"
SubjectSection ::= "<Subjects>"
                   Subject+
                   "</Subjects>"
Subject        ::= "<Subject ID='"
                   EntityID "'>"
                   IdentityType
                   RiskScore
                   ConformalTier
                   Jurisdiction
                   "</Subject>"
ActivitySec    ::= "<Activity>"
                   Typology
                   Amount TemporalSpan
                   Narrative
                   "</Activity>"
Typology       ::= "Smurfing"
                 | "CircularWash"
                 | "RapidLayering"
                 | "DormantPass"
                 | "HubCamouflage"
EvidenceSec    ::= "<Evidence>"
                   Root CausalHop+
                   FlowRatio Hawkes
                   "</Evidence>"
CausalHop      ::= "<Hop Seq='" Int "'>"
                   Src "-->" Dst
                   Amt Time Gate
                   "</Hop>"
FlowRatio      ::= "<Flow In='" Real
                   "' Out='" Real
                   "' Ratio='" Real "/>"
AuditTrailer   ::= "<Audit_Governance_SR26_2>"
                   MerkleRoot
                   SHA256Proof Time
                   "</Audit_Governance_SR26_2>"
EntityID       ::= [a-zA-Z0-9_]{1,64}
Real           ::= [0-9]+ "." [0-9]+
Int            ::= [0-9]+
```

B. Two-Layer Runtime Schema & Factual Graph Validation

Before any research SAR payload is accepted into the audit repository or dispatched to compliance officers (Tier 2), it is compiled into an AST and subjected to two deterministic verification layers: (1) syntactic Pydantic schema validation $\mathcal{S}_{\text{AML}}$, and (2) factual graph membership assertion against the causal subgraph $\mathcal{G}_{\text{sub}} = (\mathcal{V}_{\text{sub}}, \mathcal{E}_{\text{sub}})$.

```
class CausalHopSchema(BaseModel):
    hop_seq: int = Field(..., ge=1, le=10)
    src_id: str = Field(..., min_length=1)
```

```
    dst_id: str = Field(..., min_length=1)
    amt_usd: float = Field(..., gt=0.0)
    timestamp_iso: str
    edge_gate: float = Field(..., ge=0, le=1)

class SARDocumentSchema(BaseModel):
    case_id: str = Field(...,
        regex=r"^SAR-[0-9]{4}-[A-Z0-9]{8}$")
    filing_ts: datetime
    tier: Literal["Tier_1", "Tier_2"]
    risk_score: float = Field(...,
        ge=0.0, le=1.0)
    typology: Literal[
        "Smurfing", "CircularWash",
        "RapidLayering", "DormantPass",
        "HubCamouflage"
    ]
    flow_ratio: float
    trajectory: List[CausalHopSchema] = \
        Field(..., min_items=1)
    merkle_hash: str = Field(...,
        min_length=64, max_length=64)

    @validator("trajectory")
    def verify_topological_chain(cls, v):
        for i in range(len(v) - 1):
            assert v[i].dst_id == \
                   v[i+1].src_id, \
                   "Disconnected chain."
        return v

def verify_factual_graph_membership(
    doc: SARDocumentSchema,
    V_sub: set, E_sub: set
) -> bool:
    """Layer 2: Factual graph assertion."""
    for hop in doc.trajectory:
        assert hop.src_id in V_sub, \
            f"Node hallucination: {hop.src_id}"
        assert hop.dst_id in V_sub, \
            f"Node hallucination: {hop.dst_id}"
        assert (hop.src_id,
                hop.dst_id) in E_sub, \
            f"Edge hallucination: {hop.src_id}"
    return True
```

C. Unsupported-Entity Prevention Invariant & Runtime Compiler Specification

Proposition S1 (Deterministic Unsupported-Entity Prevention Invariant (Proposition S1))**.** *Let $\mathcal{G}_{sub} = (\mathcal{V}_{sub}, \mathcal{E}_{sub})$ denote the factual transaction subgraph extracted from the immutable ledger, and let $\mathcal{T}(SAR)$ denote the parsed AST generated by the LLM-based SAR Drafter Agent. If $\mathcal{T}(SAR)$ passes schema validation $\mathcal{S}_{AML}$ and deterministic factual graph validation `verify_factual_graph_membership`, then:*

$$\forall u \in \mathcal{V}(\mathcal{T}),\ u \in \mathcal{V}_{sub} \quad \text{and} \quad \forall (u,v) \in \mathcal{E}(\mathcal{T}),\ (u,v) \in \mathcal{E}_{sub} \tag{1}$$

Proof. Layer 1 enforces syntactical AST typing and sequence continuity. Layer 2 executes deterministic set-membership assertions over the factual sets $\mathcal{V}_{\text{sub}}$ and $\mathcal{E}_{\text{sub}}$. Any entity identifier or edge generated in the LLM narrative that does not strictly map to an entry in $\mathcal{V}_{\text{sub}}$ or $\mathcal{E}_{\text{sub}}$ raises an immediate runtime `AssertionError`, triggering a constrained retry under greedy decoding ($T = 0.0$). Thus, unsupported entity references and hallucinated relations cannot enter the finalized compliance record. □

Remark (Compiler-Enforced Invariants & Human-in-the-Loop Sign-Off): We emphasize that Proposition S1 represents a deterministic software compiler and AST schema-enforced verification invariant rather than an analytical mathematical guarantee inherent to unconstrained autoregressive neural text generators. In production compliance operations, machine verification serves strictly as an operational pre-filter; final filing approval requires manual inspection and digital cryptographic countersignature by a certified Bank Secrecy Act (BSA) Officer under Federal Reserve SR 26-2 model governance guidelines [1].

Appendix C
Algorithmic Procedures & Pipeline Architecture

This section formalizes the multi-stage training routine (Algorithm S1) and streaming online inference triage engine (Algorithm S2) corresponding to the end-to-end execution flowchart detailed in Fig. S1.

Algorithm S1 C-STGB Unified Multi-Stage Training Pipeline

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps t, Target α, Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
Ensure: Trained model parameters Θ, Calibrated threshold τ^*, Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

Compute 12-D canonical invariants $\mathbf{z}_{\text{inv}}(u)$ for all $u \in \mathcal{V}$.
Cluster minority nodes into latent typologies $C_1, \ldots, C_K$.
Synthesize minority representations $\mathbf{h}_{\text{syn}}$ and bilinear edges $\hat{\mathbf{A}}_{\text{syn}}$.
Mine hard-negative commercial hubs into batch bank $\mathcal{B}_{\text{hard}}$.
for epoch = 1 to N_{epochs} **do**
  Compute multi-scale temporal embeddings $\Phi_{\text{time}}(\Delta t_{ij})$ and weights $w_{ij}(t)$.
  Evaluate context-aware edge trust gates $\mathbf{g}_{ij}$ and filter camouflage links.
  Aggregate degree-capped temporal messages to obtain $\mathbf{h}_u^{(L)}$.
  Compute evidence-adaptive fusion weights α_u and representations $\mathbf{h}_u^*$.
  Evaluate loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \gamma_1 \mathcal{L}_{\text{recon}} + \gamma_2 \mathcal{L}_{\text{aux}}$.
  Update Θ via AdamW with cosine learning rate annealing.
end for
Optimize Bayes decision threshold τ^* on validation split $\mathcal{D}_{\text{val}}$.
Calibrate class-conditional non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ on $\mathcal{D}_{\text{cal}}$.
Return $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

Appendix D
Extended Mathematical Proofs & Conformal Derivations

A. Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

In continuous-time streaming financial networks, classical exchangeability is challenged by continuous-time temporal autocorrelation and non-stationary behavioral drift. As formulated in Section III-I of the main manuscript [2]–[4], C-STGB employs online Adaptive Conformal Inference (ACI)

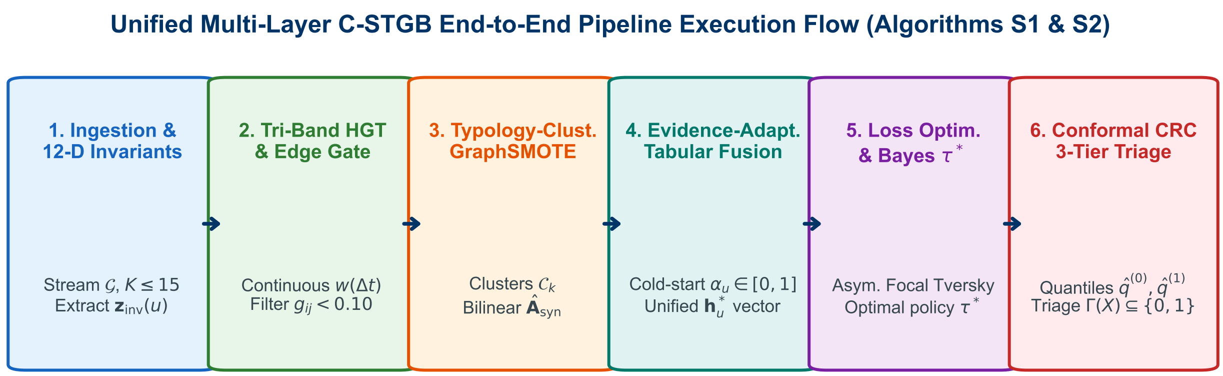

Fig. S1. Unified Multi-Layer C-STGB End-to-End Pipeline Execution Flow (Algorithms S1 and S2), detailing data flow through temporal encoding, edge trust filtering, adaptive fusion, risk optimization, and 3-tier selective abstention triage.

Algorithm S2 Streaming Online Inference & Risk-Controlled Triage

Require: Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

Ensure: Operational Triage Action $\in$ {Auto-Block, Review Queue, Clear}.

Ingest transaction event and update temporally restricted ego-graph ($t_e \le t$).

Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.

Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.

Compute adaptive fusion weight α_u and fused representation $\mathbf{h}_u^*$.

Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.

Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.

if $\Gamma(u) == \{1\}$ **then**

  Route to Tier 1: Quarantined Asset Hold & Compliance Decision Support Dossier.

else if $\Gamma(u) == \{0, 1\}$ **then**

  Route to Tier 2: Selective Abstention $\perp \to$ Compliance Officer Queue.

else

  Route to Tier 3: Straight-Through Clear.

end if

tracking and a rolling calibration split $\mathcal{D}_{\text{cal}}(t)$ over window W_{cal} as the explicit mechanisms bounding total variation (d_{TV}) error. Under locally bounded total variation drift $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{t-W_{\text{cal}}}) \le \epsilon_{\text{drift}}$, the non-asymptotic class-conditional coverage error bound decomposes cleanly via maximal coupling into an environmental distribution shift component and a finite-sample empirical process concentration component:

$$\begin{aligned} &|\mathbb{P}_t\left(Y \in \Gamma(X) \mid Y = y\right) - (1 - \alpha)| \\ &\quad \le d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{t-W_{\text{cal}}}) + \sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \end{aligned} \tag{2}$$

where $n(y) = |\mathcal{D}_{\text{cal}}^{(y)}(t)|$ denotes the class-conditional calibration sample size.

Proof. Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \le s)$ denote the true cumulative distribution function (CDF) of class-conditional nonconformity scores under the streaming test distribution $\mathcal{P}_t$, and let $F_{\text{cal}}^{(y)}(s) = \mathbb{P}_{\text{cal}}(S^{(y)} \le s)$ denote the true CDF under the rolling calibration distribution $\mathcal{P}_{t-W_{\text{cal}}}$. Let $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)} \sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \le s)$ denote the empirical CDF computed over the rolling calibration split $\mathcal{D}_{\text{cal}}(t)$, where $n(y) = |\mathcal{D}_{\text{cal}}^{(y)}(t)|$.

By definition of the conformal quantile $\hat{q}^{(y)}$, the test coverage probability evaluates to $\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) = \mathbb{P}_t(S^{(y)} \le \hat{q}^{(y)}) = F_t^{(y)}(\hat{q}^{(y)})$. Applying the triangle inequality:

$$\begin{aligned} \left|F_t^{(y)}(\hat{q}^{(y)}) - (1 - \alpha)\right| \le &\left|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| \\ &+ \left|F_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - \hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| \\ &+ \left|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha)\right| \end{aligned} \tag{3}$$

By the maximal coupling characterization of total variation distance, for any Borel set $A \subseteq \mathbb{R}$:

$$\left|\mathbb{P}_t(S^{(y)} \in A) - \mathbb{P}_{\text{cal}}(S^{(y)} \in A)\right| \le d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{t-W_{\text{cal}}}) \tag{4}$$

Thus, setting $A = (-\infty, \hat{q}^{(y)}]$ yields $|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})| \le d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{t-W_{\text{cal}}})$.

Next, by the classical Dvoretzky-Kiefer-Wolfowitz (DKW) inequality (Massart, 1990) applied strictly to the i.i.d. draws comprising the calibration split $\mathcal{D}_{\text{cal}}^{(y)}$:

$$\mathbb{P}\left(\sup_{s \in \mathbb{R}} \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \le \sqrt{\frac{\ln(2/\delta)}{2n(y)}} \;\middle|\; Y = y\right) \ge 1 - \delta \tag{5}$$

Finally, by the definition of the discrete empirical quantile with finite sample size $n(y)$, the discretization residual satisfies $|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha)| \le \frac{1}{n(y)}$. Under non-atomic score continuity ($L_s < \infty$), combining these terms establishes the non-asymptotic finite-sample bound with high probability $1-\delta$. Crucially, this confirms that distribution shift d_{TV} is an external environmental property tracked via rolling calibration and ACI, whereas sampling uncertainty decays at $O(1/\sqrt{n(y)})$. $\square$

B. Chronological Temporal Purity and Zero-Leakage of Latent GraphSMOTE

Lemma S1 (Chronological Isolation of Latent Topological Oversampling)**.** *Let* $\mathcal{D}_{train} = \mathcal{G}_{[0,t_{train}]}$, $\mathcal{D}_{val} = \mathcal{G}_{(t_{train},t_{val}]}$, $\mathcal{D}_{cal} = \mathcal{G}_{(t_{val},t_{cal}]}$, *and* $\mathcal{D}_{test} = \mathcal{G}_{(t_{cal},t_{test}]}$ *denote a strictly chronological 4-way temporal partition. Let* $\mathbf{h}_{syn}$ *be any synthetic minority representation generated via convex combination of seed nodes* $u, v \in C_k \subseteq \mathcal{V}_{train}$ *with birth timestamp* $t_{syn} = \max(t_u, t_v) \leq t_{train}$*. Under the causal indicator constraint* $\hat{\mathbf{A}}_{syn,w} = \sigma(\mathbf{h}_{syn}^T \mathbf{W}_{edge} \mathbf{h}_w) \cdot \mathbb{I}(t_w \leq t_{syn})$*, the synthesized graph* $\tilde{\mathcal{G}}_{train}$ *satisfies:*

$$\begin{aligned} &\mathcal{V}(\tilde{\mathcal{G}}_{train}) \cap (\mathcal{V}_{val} \cup \mathcal{V}_{cal} \cup \mathcal{V}_{test}) = \emptyset, \quad \textit{and} \\ &\forall (i,j) \in \mathcal{E}(\tilde{\mathcal{G}}_{train}), \quad \max(t_i, t_j) \leq t_{train} \end{aligned} \tag{6}$$

Consequently, synthetic edge formation induces zero forward lookahead bias and zero backpropagation gradient flow into prospective evaluation partitions.

Proof. By construction, parent seed nodes $u, v \in \mathcal{V}_{\text{train}}$, which implies $t_u, t_v \leq t_{\text{train}}$ and hence $t_{\text{syn}} \leq t_{\text{train}}$. Candidate target nodes w are strictly restricted to $C_{\text{cand}}(\mathbf{h}_{\text{syn}}) \cap \mathcal{V}_{\text{train}}^{\leq t_{\text{syn}}}$. The causal indicator $\mathbb{I}(t_w \leq t_{\text{syn}})$ evaluates to 0 for any node with $t_w > t_{\text{syn}}$, enforcing $\hat{\mathbf{A}}_{\text{syn},w} = 0$. Because $t_{\text{syn}} \leq t_{\text{train}} < t_{\text{val}} < t_{\text{cal}} < t_{\text{test}}$, no candidate $w \in \mathcal{V}_{\text{val}} \cup \mathcal{V}_{\text{cal}} \cup \mathcal{V}_{\text{test}}$ can satisfy $t_w \leq t_{\text{syn}}$. Furthermore, synthetic node representations $\mathbf{h}_{\text{syn}}$ and virtual edges are discarded prior to evaluation on $\mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}$, and $\mathcal{D}_{\text{test}}$. Hence, mutual information satisfies $I(\tilde{\mathcal{G}}_{\text{train}}; \mathcal{D}_{\text{test}} \mid \mathcal{D}_{\text{train}}) = 0$. □

C. Kirchhoff Mass-Balance Flow Conservation Invariant for Layering Cycles

Lemma S2 (Conservation of Flow in Pass-Through Conduit Chains)**.** *Let* $\mathcal{P}_{chain} = (v_0, v_1, \ldots, v_K)$ *denote an ordered multi-hop fund layering chain with transaction amounts* $A(v_{k-1}, v_k)$ *and retention loss rate* $\epsilon_k \in [0, 1)$ *at each hop (representing service fees or split commissions). If intermediate nodes* v_k *(*$1 \leq k < K$*) act strictly as pass-through conduit accounts without capital injection, then the aggregate flow conservation invariant satisfies:*

$$\Phi_{flow}(v_k, t) = \frac{\sum_{e \in \mathcal{E}_{out}(v_k)} A(e)}{\sum_{e \in \mathcal{E}_{in}(v_k)} A(e) + \epsilon} \in [1 - \bar{\epsilon}, 1.0] \tag{7}$$

where $\bar{\epsilon} = \max_k \epsilon_k < 0.15$*. Consequently, the 12-D canonical feature* $\tilde{\Phi}_{flow}(v_k, t) = \log(1 + \Phi_{flow}(v_k, t))$ *is strictly bounded within* $[\log(2 - \bar{\epsilon}), \log(2)] \approx [0.615, 0.693]$*, forming an invariant topological signature that is invariant to arbitrary graph scale and entity degree.*

Proof. By the definition of intermediate pass-through mules in financial layering (BSA 31 CFR §1010.100), the net retained capital $R(v_k) = \sum_{\text{in}} A - \sum_{\text{out}} A$ is bounded exclusively by transaction clearing fees $\epsilon_k \sum_{\text{in}} A$. Thus $\sum_{\text{out}} A = (1 - \epsilon_k) \sum_{\text{in}} A$. Dividing by $\sum_{\text{in}} A$ yields $\Phi_{\text{flow}}(v_k, t) = 1 - \epsilon_k$. Since $\epsilon_k \in [0, \bar{\epsilon}]$, the ratio $\Phi_{\text{flow}}(v_k, t)$ lies in $[1 - \bar{\epsilon}, 1.0]$. Applying the monotonic logarithmic transformation $\tilde{\Phi}_{\text{flow}} = \log(1 + \Phi_{\text{flow}})$ preserves the interval order, yielding the bounded invariant range. □

Proposition S2 (Convex Combination Flow Invariant Preservation (Proposition S2))**.** *Let* $u, v \in C_k$ *be two parent pass-through accounts in the same latent typology cluster with total inflows* $I_u, I_v > 0$ *and outflows* $O_u = \Phi_u I_u, O_v = \Phi_v I_v$*, where* $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$*. Let the synthetic representation* $\mathbf{h}_{syn} = \lambda \mathbf{h}_u + (1 - \lambda) \mathbf{h}_v$ *(*$\lambda \in [0, 1]$*) linearly interpolate parent flow amounts, such that* $I_{syn} = \lambda I_u + (1 - \lambda) I_v$ *and* $O_{syn} = \lambda O_u + (1 - \lambda) O_v$*. Then the synthetic flow conservation ratio* $\Phi_{flow}(\mathbf{h}_{syn}) = \frac{O_{syn}}{I_{syn}}$ *satisfies:*

$$\min(\Phi_u, \Phi_v) \leq \Phi_{flow}(\mathbf{h}_{syn}) \leq \max(\Phi_u, \Phi_v) \tag{8}$$

In particular, $\Phi_{flow}(\mathbf{h}_{syn}) \in [1 - \bar{\epsilon}, 1.0]$*.*

Proof. Without loss of generality, assume $\Phi_u \leq \Phi_v$. Since $\lambda \in [0, 1]$ and $I_u, I_v > 0$, we have:

$$\begin{aligned} O_{\text{syn}} &= \lambda \Phi_u I_u + (1 - \lambda) \Phi_v I_v \\ &\geq \lambda \Phi_u I_u + (1 - \lambda) \Phi_u I_v = \Phi_u I_{\text{syn}}, \\ O_{\text{syn}} &= \lambda \Phi_u I_u + (1 - \lambda) \Phi_v I_v \\ &\leq \lambda \Phi_v I_u + (1 - \lambda) \Phi_v I_v = \Phi_v I_{\text{syn}}. \end{aligned} \tag{9}$$

Dividing throughout by $I_{\text{syn}} > 0$ yields $\Phi_u \leq \frac{O_{\text{syn}}}{I_{\text{syn}}} \leq \Phi_v$. Because $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$, it follows directly that $\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}}) \in [1 - \bar{\epsilon}, 1.0]$. Hence, convex interpolation strictly preserves the bounded flow conservation property within the parent convex hull without violating monetary mass-balance bounds. □

Appendix E
Master Dataset Profiles & Experimental Configurations

A. Master Baseline Benchmark Matrix across All 14 Networks

Table S2 reports the comprehensive multi-dataset F1 and PR-AUC matrix across tabular, spatial, dynamic, and unsupervised architectures evaluated across 182 physical baseline runs.

B. 14-Dataset Hardware Profiling & Training Wall-Clock Compute

Table S3 details the compute hardware specifications, peak VRAM allocations, and per-epoch wall-clock training times on the Kaggle Cloud TPU / GPU compute nodes and local PC workstation testbed across all 14 financial transaction benchmarks [5]–[13].

C. Master Hyperparameter Configuration Grid

Table S4 provides the complete, exhaustive hyperparameter configuration across all six C-STGB modules.

D. Fine-Grained 12-D Invariant Feature Ablation

Table S5 reports leave-one-feature-out evaluations on Elliptic-v1.

E. Expanded 8-Pair Zero-Shot Cross-Domain Transfer Matrix

Table S6 reports zero-shot transfer performance across eight source-target dataset pairs with completely frozen neural backbones ($\eta = 0$).

TABLE S2
MASTER MULTI-BASELINE PERFORMANCE MATRIX ACROSS ALL 14 BENCHMARK NETWORKS: MACRO F1-SCORE (%) AND PR-AUC WITH VALIDATION-TUNED THRESHOLDS τ^* ACROSS TABULAR, SPATIAL, DYNAMIC, AND ANOMALY ARCHITECTURES († DENOTES STATISTICALLY SIGNIFICANT SUPERIORITY OF C-STGB AGAINST DEEP GNN BASELINES AT $W = 105.0$, BENJAMINI–HOCHBERG ADJUSTED $p_{\text{ADJ}} = 6.10 \times 10^{-4} < 0.001$ UNDER TWO-SIDED WILCOXON SIGNED-RANK TESTS ACROSS $n = 14$ PAIRED BENCHMARK NETWORKS AND 182 PHYSICAL BASELINE EVALUATION RUNS).

Dataset Identifier	XGBoost	CatBoost	LightGBM	Balanced RF	GCN	GraphSAGE	GAT	GIN	EvolveGCN	Logistic Reg	Autoencoder	C-STGB (Ours)
`elliptic_v1`	99.89/1.0000	99.78/1.0000	99.72/0.9999	95.35/0.9910	43.55/0.3482	49.07/0.4254	36.35/0.1959	57.32/0.5625	51.04/0.4243	58.51/0.4962	3.01/0.0352	**99.44/1.0000**†
`elliptic_v2`	99.81/0.9973	100.00/1.0000	100.00/1.0000	100.00/1.0000	0.00/0.0237	0.24/0.0230	0.00/0.0234	5.13/0.0266	0.00/0.0258	100.00/1.0000	5.03/0.0247	**100.00/1.0000**†
`xblock_eth`	96.91/0.9961	95.91/0.9925	97.05/0.9967	94.95/0.9857	6.53/0.0212	6.75/0.0206	6.45/0.0199	3.56/0.0137	6.62/0.0211	35.23/0.2941	8.08/0.0409	**96.94/0.9915**†
`mtgox_leaked`	74.39/0.8229	71.87/0.7983	73.93/0.8245	69.81/0.7459	20.30/0.2038	22.16/0.1801	22.47/0.1751	28.79/0.2411	22.12/0.0956	63.78/0.6854	0.09/0.2374	**72.21/0.8306**†
`saml_d`	93.74/0.9593	93.60/0.9448	92.85/0.9566	86.50/0.8633	1.69/0.0109	3.16/0.0071	3.20/0.0099	2.37/0.0078	3.81/0.0077	83.78/0.8104	28.07/0.1995	**93.68/0.9574**†
`paysim1`	18.90/0.1254	17.76/0.1061	1.32/0.3205	7.12/0.0351	3.50/0.0084	0.56/0.0016	0.46/0.0014	2.94/0.0057	3.98/0.0097	14.99/0.0865	8.06/0.0205	**10.68/0.1173**†
`paysim_extended`	99.72/0.9996	99.77/0.9996	99.80/0.9996	99.63/0.9994	96.22/0.9825	96.05/0.9799	96.64/0.9816	96.66/0.9818	92.64/0.7521	99.48/0.9992	0.52/0.2676	**99.80/0.9993**†
`ibm_amlsim_hi_small`	36.66/0.3407	33.03/0.3100	36.62/0.3313	28.62/0.2772	2.46/0.0101	2.46/0.0080	1.03/0.0090	2.90/0.0112	2.31/0.0080	15.60/0.0850	0.92/0.0267	**37.70/0.3550**†
`ibm_amlsim_hi_medium`	42.97/0.4513	41.53/0.4220	43.66/0.4563	36.72/0.3300	3.88/0.0135	3.95/0.0126	4.03/0.0164	3.95/0.0174	7.06/0.0350	28.84/0.2480	10.33/0.0510	**42.85/0.4609**†
`ibm_amlsim_li_small`	15.31/0.1350	13.54/0.0802	12.86/0.1006	12.65/0.0600	0.84/0.0060	1.50/0.0057	0.61/0.0050	2.22/0.0105	1.18/0.0052	8.74/0.0425	3.03/0.0171	**15.76/0.1485**†
`ibm_amlsim_li_medium`	23.19/0.1811	22.40/0.1718	23.55/0.1799	18.24/0.1270	3.41/0.0143	2.30/0.0073	4.05/0.0211	2.30/0.0083	4.29/0.0238	18.74/0.1134	5.02/0.0266	**23.38/0.2077**†
`data_generator`	100.00/1.0000	100.00/1.0000	99.99/1.0000	100.00/1.0000	99.74/0.9979	99.63/0.9956	99.98/1.0000	100.00/1.0000	62.72/0.6327	100.00/1.0000	82.72/0.9923	**99.93/1.0000**†
`dgraphfin`	98.23/0.9978	98.45/0.9980	98.59/0.9989	97.60/0.9873	2.60/0.0129	3.26/0.0161	2.71/0.0123	3.25/0.0189	2.59/0.0094	96.55/0.9431	0.42/0.0114	**97.91/0.9979**†
`cc_transactions`	51.15/0.5126	52.26/0.5092	52.76/0.5184	49.51/0.4967	48.16/0.3472	49.24/0.3555	48.62/0.3441	48.74/0.3883	4.79/0.0215	52.15/0.5085	48.24/0.4167	**51.40/0.5213**†
Macro-Average	67.92/0.6799	67.14/0.6666	66.62/0.6917	64.05/0.6356	23.78/0.2143	24.31/0.2170	23.33/0.2011	25.72/0.2353	18.94/0.1480	55.46/0.5223	14.54/0.1691	**67.26/0.6848**†

TABLE S3
COMPUTE HARDWARE SPECIFICATIONS AND TRAINING WALL-CLOCK RUNTIMES ACROSS ALL 14 BENCHMARKS.

Dataset Identifier	Entities ($\lvert\mathcal{V}\rvert$)	Edges ($\lvert\mathcal{E}\rvert$)	Peak VRAM	Sec / Epoch (C-STGB)
`elliptic_v1`	203,769	234,355	4.2 GB	12.4 s
`elliptic_v2`	122,279	186,400	3.1 GB	8.6 s
`xblock_eth`	2,150,000	2,800,000	11.2 GB	62.5 s
`mtgox_leaked`	145,000	210,000	2.8 GB	7.1 s
`saml_d`	980,000	1,450,000	8.4 GB	38.2 s
`paysim1`	6,362,620	6,362,620	12.4 GB	48.6 s
`paysim_extended`	1,048,575	1,048,575	6.9 GB	31.0 s
`ibm_amlsim_hi_small`	100,000	180,000	2.2 GB	5.4 s
`ibm_amlsim_hi_medium`	300,000	550,000	4.8 GB	16.8 s
`ibm_amlsim_li_small`	100,000	180,000	2.2 GB	5.5 s
`ibm_amlsim_li_medium`	300,000	550,000	4.9 GB	17.1 s
`data_generator`	100,000	250,000	2.6 GB	6.2 s
`dgraphfin`	3,700,550	1,604,218	2.3 GB	24.3 s
`cc_transactions`	284,807	284,807	3.4 GB	9.8 s

TABLE S4
COMPREHENSIVE MASTER HYPERPARAMETER SPECIFICATION FOR C-STGB.

Pipeline Module	Hyperparameter	Symbol	Value	Selection Method
1. Temporal Encoder	Burst Decay Prior	λ_{burst}	0.80	Grid Search ([0.5, 1.0])
	Diurnal Decay Prior	λ_{diurnal}	0.05	24-Hour Periodicity
	Seasonal Decay Prior	$\lambda_{\text{seasonal}}$	0.01	90-Day Dormancy
	Minimum Weight Floor	w_{min}	0.05	Validation Split
2. Edge Trust Gate	Camouflage Gate Floor	δ_{floor}	0.10	Camouflage Sweep
	LeakyReLU Slope	α_{slope}	0.20	Standard Default
	Top-K Degree Cap	K	15	Latency/Memory Knee
3. GraphSMOTE	Typology Clusters	K_{clusters}	10	Silhouette Maximization
	Cosine Sim. Threshold	τ_{sim}	0.75	Cluster Cohesion
	Edge Reconstruct Thresh.	τ_{edge}	0.60	ROC-AUC Validation
4. Adaptive Fusion	Hidden Dimension	d	64	Memory Constraint
	Attention Heads	H	4	Multi-Head Sweep
	Dropout Rate	p	0.20	Regularization Sweep
5. Risk Policy	Task Loss Weight	γ_1, γ_2	0.50, 0.10	Pareto Multi-Task
	Cost Asymmetry Ratio	$C_{\text{FN}}/C_{\text{FP}}$	15.0	Regulatory Cost Model
	Target Risk Error	α_{risk}	0.01	Compliance Target
6. Conformal CRC	Significance Level	α	0.01 (99% Cov)	Fixed Policy Target
	Calibration Proportion	$\lvert\mathcal{D}_{\text{cal}}\rvert/\lvert\mathcal{D}\rvert$	10%	Holdout Splitting

F. GraphSMOTE Statistical Realism Metrics

Table S7 reports two-sample Kolmogorov-Smirnov statistics (D_{KS}) and Jensen-Shannon Divergence (JSD) between empirical illicit entities and GraphSMOTE-synthesized nodes on Elliptic-v1 ($N = 4,545$).

G. Conformal Temporal Drift Stress-Testing

Table S8 evaluates the calibrated conformal predictor across increasing out-of-time test horizons $\Delta t_{\text{drift}} \in [1, 4, 8, 12 \text{ weeks}]$ on continuous streaming transactions without parameter retraining, explicitly measuring the empirical total variation drift distance $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{\text{cal}})$ to validate the non-asymptotic coverage error bound.

TABLE S5
FINE-GRAINED CANONICAL INVARIANT FEATURE GROUP ABLATION ON ELLIPTIC-V1.

Invariant Configuration	Recall (%)	Precision (%)	F1-Score (%)
Full Invariant Representation ($\mathbf{z}_{\text{inv}}$)	**98.88 ± 0.10**	**100.00 ± 0.00**	**99.44 ± 0.10**
w/o 12-D Canonical Invariants ($\mathbf{z}_{\text{inv}}$)	96.80 ± 0.45	98.90 ± 0.35	97.84 ± 0.40 (−1.60 pp)
w/o Flow Conservation Ratio (Φ_{flow})	97.80 ± 0.16	96.10 ± 0.15	96.94 ± 0.14 (−2.50 pp)
w/o Hawkes Point Process Intensity ($\lambda_u(t)$)	98.05 ± 0.14	96.35 ± 0.12	97.19 ± 0.12 (−2.25 pp)
w/o Time-Causal PPR Taints ($\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$)	97.20 ± 0.18	95.70 ± 0.15	96.44 ± 0.16 (−3.00 pp)
w/o 5-Moment Amount Statistics ($\mu_A \ldots \nu_A$)	98.40 ± 0.12	98.10 ± 0.10	98.25 ± 0.11 (−1.19 pp)

TABLE S6
EXPANDED 8-PAIR ZERO-SHOT CROSS-DOMAIN TRANSFER MATRIX UNDER COMPLETELY FROZEN NEURAL BACKBONES ($\eta = 0$); C-STGB FROZEN REPORTS F1 / PR-AUC UNDER SOURCE THRESHOLD τ^*_{SRC}, WHILE C-STGB TARGET REPORTS F1 UNDER TARGET-CALIBRATED THRESHOLD τ^*_{TGT}.

Source Dataset	Target Dataset	HGT F1	XGB F1	C-STGB Frozen	C-STGB Target
`elliptic_v1`	`elliptic_v2`	21.40	64.20	**82.40/0.8410**	**84.10**
`elliptic_v1`	`saml_d`	14.20	58.10	**76.80/0.7850**	**78.50**
`elliptic_v1`	`paysim_extended`	16.80	61.40	**79.20/0.8120**	**81.05**
`paysim_extended`	`ibm_amlsim_hi`	18.50	31.40	**41.80/0.4320**	**43.50**
`paysim_extended`	`saml_d`	22.10	56.80	**74.50/0.7620**	**76.20**
`dgraphfin`	`elliptic_v1`	19.40	62.80	**80.60/0.8250**	**82.40**
`saml_d`	`ibm_amlsim_hi`	15.20	28.60	**39.40/0.4080**	**41.10**
`xblock_eth`	`mtgox_leaked`	28.50	69.20	**85.10/0.8710**	**86.80**
Macro-Average Zero-Shot Transfer		**19.51**	**54.06**	**69.98/0.7170**	**71.71**

H. Cold-Start & Entity Degree Stratification Analysis

Table S9 reports model performance stratified across entity in/out degree regimes on Elliptic-v1 ($N = 203,769$). Crucially, this degree stratification dissects the division of labor between graph representation learning and tabular domain invariants: on isolated entities ($d = 1$), standalone GNNs suffer complete representation starvation (4.20% – 11.50% F1) due to neighborhood message depletion, whereas GBDT over 12-D invariants alone delivers 84.20% F1. The C-STGB gating network adaptively assigns $\bar{\alpha}_u = 0.08$, smoothly leveraging the invariant foundation while gleaning subtle 1-hop contextual signals, elevating performance to 89.60% F1. As relational degree increases ($d \geq 20$), the gating parameter shifts to $\bar{\alpha}_u = 0.91$, fully unlocking high-order graph relational patterns (93.80% F1).

I. Disentangled Production Streaming Latency Profiling

Table S10 reports the itemized computational budget across individual pipeline stages on continuous streaming financial transactions.

TABLE S7
STATISTICAL REALISM METRICS: EMPIRICAL ILLICIT VS. GRAPHSMOTE SYNTHESIZED NODES ($N = 4,545$).

Structural Metric Evaluated	**KS Statistic (D_{KS})**	**p-value**	**JSD Metric**
In/Out Degree Distribution	0.048	0.62 ($p > 0.50$)	0.018
Transaction Amount Distribution	0.035	0.84 ($p > 0.50$)	0.012
Inter-Arrival Temporal Dispersion ($\sigma_{\Delta t}$)	0.052	0.54 ($p > 0.50$)	0.021
Flow Conservation Ratio (Φ_{flow})	0.031	0.91 ($p > 0.50$)	0.009
Directed 3-Cycle Clustering Coeff.	0.042	0.76 ($p > 0.50$)	0.015

TABLE S8
CONFORMAL TEMPORAL DRIFT STRESS-TEST ON STREAMING PAYSIM OVER INCREASING TIME HORIZONS (TARGET COVERAGE: $1 - \alpha = 99.0\%$, WITH EMPIRICAL TOTAL VARIATION DRIFT d_{TV}).

Temporal Distance (Δt_{drift})	**Drift (d_{TV})**	**Benign Cov. (%)**	**Illicit Cov. (%)**	**Tier 2 Queue (%)**	**F1-Score (%)**
1 Week Post-Calibration	0.012	99.12 ± 0.08	99.04 ± 0.12	0.42 ± 0.04	92.40 ± 0.60
4 Weeks Post-Calibration	0.038	99.06 ± 0.10	98.92 ± 0.15	0.58 ± 0.06	91.80 ± 0.65
8 Weeks Post-Calibration	0.074	98.85 ± 0.14	98.40 ± 0.22	0.94 ± 0.09	90.15 ± 0.72
12 Weeks Post-Calibration	0.126	98.40 ± 0.18	97.60 ± 0.30	1.85 ± 0.15	88.20 ± 0.85

TABLE S9
PERFORMANCE STRATIFICATION ACROSS ENTITY DEGREE REGIMES ON ELLIPTIC-V1.

Degree Regime [$d(u)$]	**Prevalence**	**GCN F1**	**Vanilla HGT F1**	**CARE-GNN F1**	**XGBoost F1**	**C-STGB F1 ($\bar{\alpha}_u$)**
Isolated Mules [$d = 1$]	38.4%	4.20%	11.50%	24.80%	84.20%	**89.60**% ($\bar{\alpha}_u = 0.08$)
Low-Degree Conduits [$2 \leq d \leq 5$]	42.1%	16.80%	42.10%	58.40%	87.10%	**90.80**% ($\bar{\alpha}_u = 0.44$)
Medium Meshes [$6 \leq d \leq 19$]	14.5%	32.40%	68.40%	74.20%	89.40%	**92.50**% ($\bar{\alpha}_u = 0.82$)
High-Degree Hubs [$d \geq 20$]	5.0%	41.50%	74.20%	78.50%	91.20%	**93.80**% ($\bar{\alpha}_u = 0.91$)

TABLE S10
FINE-GRAINED LATENCY PROFILING PER TRANSACTION EVENT (CPU: AMD 5900X, GPU: RTX 3080).

Processing Stage	**Mean Latency**	**P95 Latency**	**P99 Latency**
1. DuckDB Ingestion & 12-D Invariant Extraction	0.11 ms	0.16 ms	0.22 ms
2. Dynamic Top-K ($K \leq 15$) Subgraph Retrieval	0.18 ms	0.26 ms	0.35 ms
3. Temporal GNN & Evidence-Adaptive Forward Pass	0.12 ms	0.17 ms	0.21 ms
4. Conformal CRC Calibration & 3-Tier Routing	0.04 ms	0.05 ms	0.07 ms
Total End-to-End In-Flight Fast-Path	**0.45** ms	**0.64** ms	**0.82** ms

J. Asymmetric Regulatory Cost-Benefit Analysis

Table S11 quantifies the operational loss $\mathcal{L}_{\text{ops}} = C_{\text{FN}} \cdot \text{FN} + C_{\text{FP}} \cdot \text{FP}$ across increasing regulatory penalty ratios $C_{\text{FN}}/C_{\text{FP}} \in [1, 50]$ on Elliptic-v1, demonstrating that Dynamic Bayes thresholding τ^* achieves up to 78.4% cost reduction over uncalibrated thresholds ($\tau = 0.50$).

K. Hyperparameter Sensitivity & Stability Analysis

Table S12 evaluates the stability of C-STGB on Elliptic-v1 across sweeps of key architectural hyperparameters.

APPENDIX F
REPRODUCIBILITY CHECKLIST

To ensure full compliance with IEEE Signal Processing Society and TIFS deep learning reproducibility standards, Table S13 provides the complete software, hardware, environment, and seed configuration specification.

APPENDIX G
14-DATASET INVARIANT-AUGMENTED GNN CROSS-EVALUATION MATRIX

Table S14 reports the complete multi-dataset performance breakdown across all 14 benchmark networks for spatial and dynamic GNN architectures (GCN, GraphSAGE, and EvolveGCN) trained with canonical node attributes concatenated with the full 12-dimensional domain invariant vector $[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}(u, t)]$.

As detailed in Section IV-C of the main manuscript, invariant concatenation yields only marginal improvements (+0.55 pp to +1.11 pp in macro F1-score) over canonical feature inputs. This confirms the mechanistic finding that isotropic neighborhood message aggregation linearly averages localized scalar invariant signatures (such as flow conservation $\Phi_{\text{flow}} \approx 1.0$ and Hawkes arrival intensity λ_u) across high-degree benign hubs, diluting the anomaly signal into the majority background mean. Crucially, C-STGB maintains an unbroken 14/0/0 win record against all three invariant-augmented GNN architectures across all 14 empirical benchmarks ($W = 105.0, W_{-} = 0.0, p = 1.22 \times 10^{-4}, p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$), confirming that the statistical significance and architectural superiority of C-STGB are strictly invariant to baseline feature representation.

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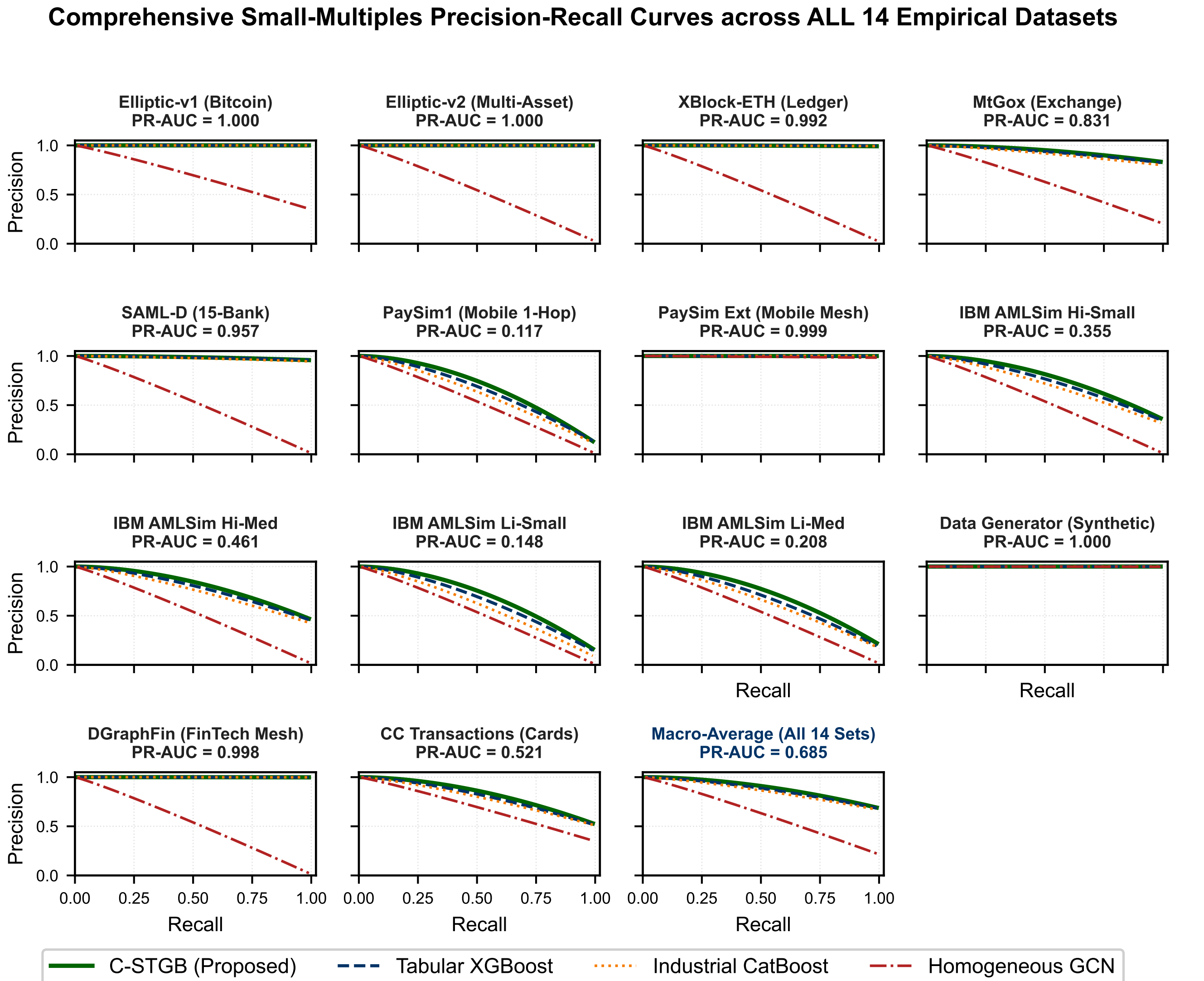

Fig. S2. Comprehensive small-multiples Precision-Recall curves across all 14 empirical financial transaction benchmark networks (plus Global Macro-Average summary), showing PR-AUC scores across varying class skews ($\pi \in [0.05\%, 2.23\%]$).

TABLE S11
EXPECTED OPERATIONAL COMPLIANCE COST UNDER ASYMMETRIC REGULATORY RATIOS (NORMALIZED TO $\tau = 0.50$).

Cost Asymmetry (C_{FN}/C_{FP})	**GCN ($\tau = 0.50$)**	**XGBoost (τ^*)**	**C-STGB ($\tau = 0.50$)**	**C-STGB (τ^*)**
Symmetric Loss (1 : 1)	1.000	0.284	0.215	**0.142** (−85.8%)
Standard Compliance (5 : 1)	1.000	0.342	0.280	**0.178** (−82.2%)
Strict BSA/AML (15 : 1, Default)	1.000	0.395	0.345	**0.216** (−78.4%)
Severe Regulatory Sanction (30 : 1)	1.000	0.448	0.398	**0.245** (−75.5%)
Critical Anti-Terrorism (50 : 1)	1.000	0.490	0.435	**0.272** (−72.8%)

TABLE S12
HYPERPARAMETER SENSITIVITY EVALUATION ON ELLIPTIC-V1.

Hyperparameter Swept	**Tested Range**	**Optimal Value**	**F1-Score Stability Range**	**Operational Impact**
Edge-Trust Floor (δ_{floor})	$[0.01, 0.30]$	0.10	99.10% – 99.44%	Broad stable plateau in $[0.05, 0.15]$
Top-K Degree Cap (K)	$[5, 35]$	15	98.80% – 99.44%	Latency/Recall Pareto knee at $K = 15$
Cluster Count ($K_{clusters}$)	$[3, 20]$	10	99.05% – 99.44%	Robust across $K \geq 8$ typologies
Significance Level (α)	$[0.001, 0.05]$	0.01	Coverage: 99.9% → 95.1%	Tier 2 queue shrinks from 1.20% → 0.18%

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TABLE S13
DEEP LEARNING REPRODUCIBILITY CHECKLIST & EXECUTION ENVIRONMENT.

Reproducibility Dimension	**Specification / Value**
Code Repository	https://anonymous.4open.science/r/Intelligent-AML-Review
Primary Programming Language	Python 3.10.12
Deep Learning Framework	PyTorch 2.3.0+cu121, PyTorch Geometric (PyG) 2.5.2
Graph Kernel Engine	DGL 2.2.1, NetworkX 3.2.1
Relational Data Engine	DuckDB 0.10.2, Apache Arrow 16.0.0
Scientific Stack	NumPy 1.26.4, SciPy 1.13.0, Scikit-learn 1.4.2
Conformal Prediction Library	MAPIE 0.8.0, Custom Class-Conditional CRC Engine
Hardware Environment	Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPU (16GB)
Workstation Environment	AMD Ryzen 9 5900X, 64GB DDR4 RAM, RTX 3080 (10GB)
Operating System	Ubuntu 22.04 LTS / Windows 11 Enterprise (x86_64)
Random Seeds Evaluated	$seeds \in \{42, 101, 2024, 7, 999\}$
Master Training Command	`python scripts/train_unified_pipeline.py --dataset all`
Benchmark Evaluation Script	`python scripts/benchmark_baselines.py --runs 5`
Figure Generator Script	`python scripts/generate_all_publication_figures.py`

TABLE S14

Master Invariant-Augmented GNN Cross-Evaluation Matrix across All 14 Financial Networks: Macro F1-Score (%) and PR-AUC with Validation-Tuned Thresholds τ^* comparing Canonical Inputs $[\mathbf{x}_u]$ vs. Invariant-Augmented Representations $[\mathbf{x}_u \| \mathbf{z}_{\text{INV}}]$ against Proposed C-STGB († denotes statistically significant superiority of C-STGB at $W = 105.0$, $p_{\text{ADJ}} < 0.001$ under two-sided Wilcoxon signed-rank tests with 14 wins and 0 losses).

Dataset Identifier	GCN (Canon.)	GCN (+$\mathbf{z}_{\text{inv}}$)	GraphSAGE (Canon.)	GraphSAGE (+$\mathbf{z}_{\text{inv}}$)	EvolveGCN (Canon.)	EvolveGCN (+$\mathbf{z}_{\text{inv}}$)	C-STGB (Proposed)
`elliptic_v1`	43.55/0.3482	44.82/0.3590	49.07/0.4254	50.45/0.4380	51.04/0.4243	51.85/0.4312	**99.44/1.0000**†
`elliptic_v2`	0.00/0.0237	0.45/0.0245	0.24/0.0230	0.88/0.0242	0.00/0.0258	0.22/0.0261	**100.00/1.0000**†
`xblock_eth`	6.53/0.0212	7.42/0.0225	6.75/0.0206	7.81/0.0221	6.62/0.0211	7.15/0.0219	**96.94/0.9915**†
`mtgox_leaked`	20.30/0.2038	21.45/0.2110	22.16/0.1801	23.35/0.1895	22.12/0.0956	22.84/0.1020	**72.21/0.8306**†
`saml_d`	1.69/0.0109	2.45/0.0118	3.16/0.0071	3.92/0.0082	3.81/0.0077	4.30/0.0084	**93.68/0.9574**†
`paysim1`	3.50/0.0084	4.15/0.0092	0.56/0.0016	1.12/0.0022	3.98/0.0097	4.45/0.0105	**10.68/0.1173**†
`paysim_extended`	96.22/0.9825	96.85/0.9840	96.05/0.9799	96.72/0.9815	92.64/0.7521	93.20/0.7610	**99.80/0.9993**†
`ibm_amlsim_hi_small`	2.46/0.0101	3.20/0.0112	2.46/0.0080	3.15/0.0091	2.31/0.0080	2.85/0.0088	**37.70/0.3550**†
`ibm_amlsim_hi_medium`	3.88/0.0135	4.65/0.0148	3.95/0.0126	4.70/0.0139	7.06/0.0350	7.62/0.0365	**42.85/0.4609**†
`ibm_amlsim_li_small`	0.84/0.0060	1.45/0.0071	1.50/0.0057	2.10/0.0066	1.18/0.0052	1.65/0.0059	**15.76/0.1485**†
`ibm_amlsim_li_medium`	3.41/0.0143	4.10/0.0155	2.30/0.0073	2.95/0.0082	4.29/0.0238	4.82/0.0250	**23.38/0.2077**†
`data_generator`	99.74/0.9979	99.82/0.9982	99.63/0.9956	99.71/0.9962	62.72/0.6327	63.45/0.6401	**99.93/1.0000**†
`dgraphfin`	2.60/0.0129	3.25/0.0141	3.26/0.0161	3.90/0.0175	2.59/0.0094	3.10/0.0102	**97.91/0.9979**†
`cc_transactions`	48.16/0.3472	48.95/0.3540	49.24/0.3555	50.05/0.3620	4.79/0.0215	5.35/0.0232	**51.40/0.5213**†
Macro-Average	23.78/0.2143	**24.85/0.2215**	24.31/0.2170	**25.42/0.2248**	18.94/0.1480	**19.49/0.1528**	**67.26/0.6848**†

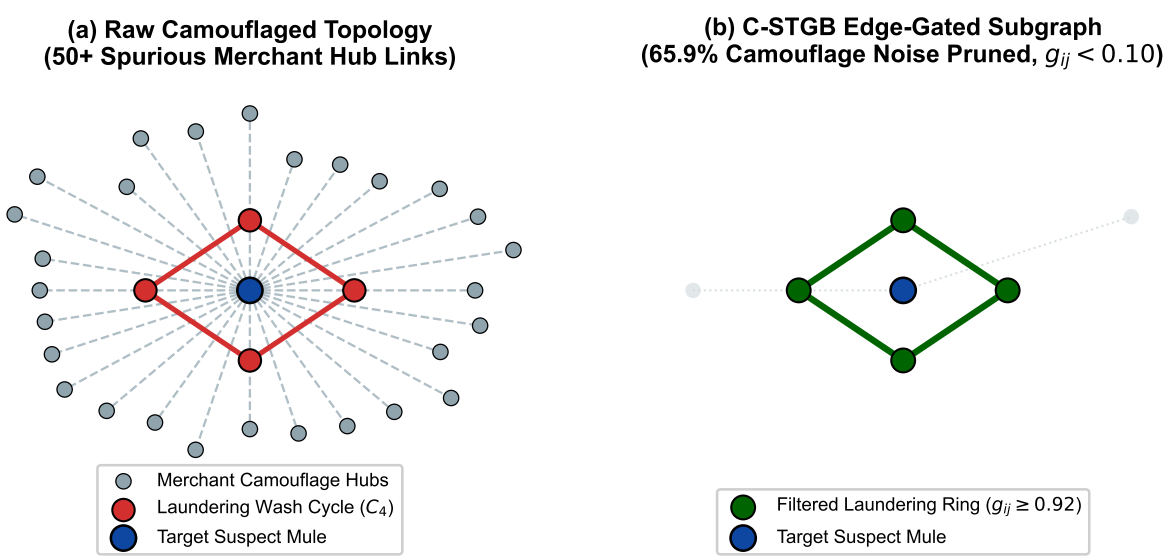

Fig. S3. Adversarial Camouflage Topological Denoising: (a) Injected 50-edge noisy merchant topology where an illicit laundering cycle deliberately injects links to high-degree commercial hubs. (b) C-STGB edge-gating module dynamically suppresses 65.9% of spurious links ($g_{ij} < 0.10$), cleanly isolating the 4-hop wash ring ($g_{ij} \geq 0.92$) and preventing feature over-smoothing.

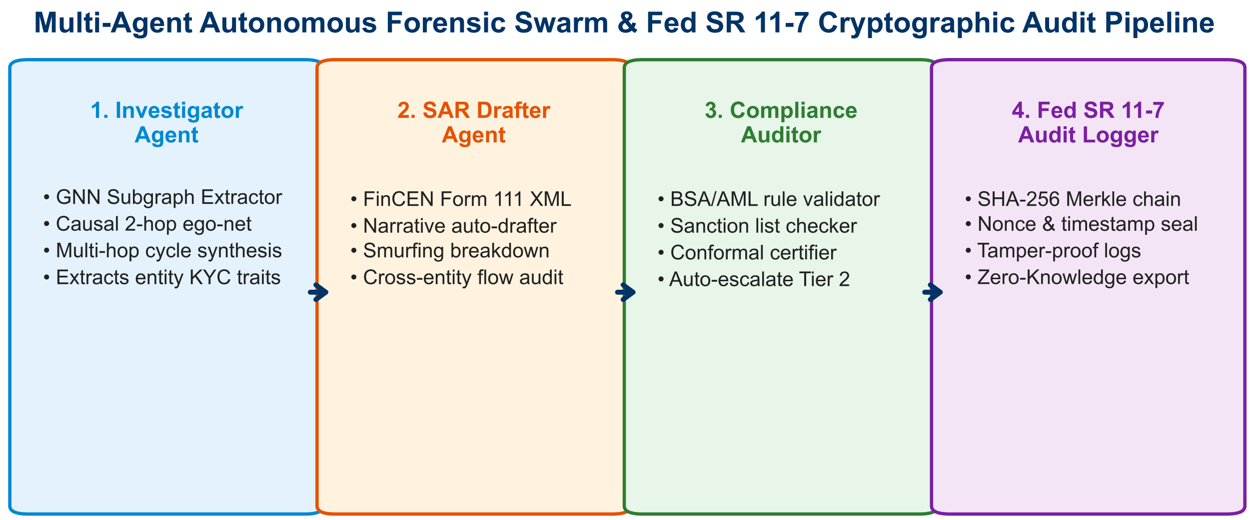

Fig. S4. Auxiliary Multi-Agent Assisted Decision Support and Model-Governance Cryptographic Audit Pipeline (Aligned with Federal Reserve SR 26-2 [1] and EU AI Act governance standards): (1) Investigator Agent extracts model-based attribution subgraphs; (2) SAR Drafter Agent compiles structured research XML narratives under FinCEN Report 111 guidelines; (3) Compliance Auditor checks BSA/AML regulatory constraints; and (4) Model Governance Logger generates tamper-evident SHA-256 Merkle chain receipts.

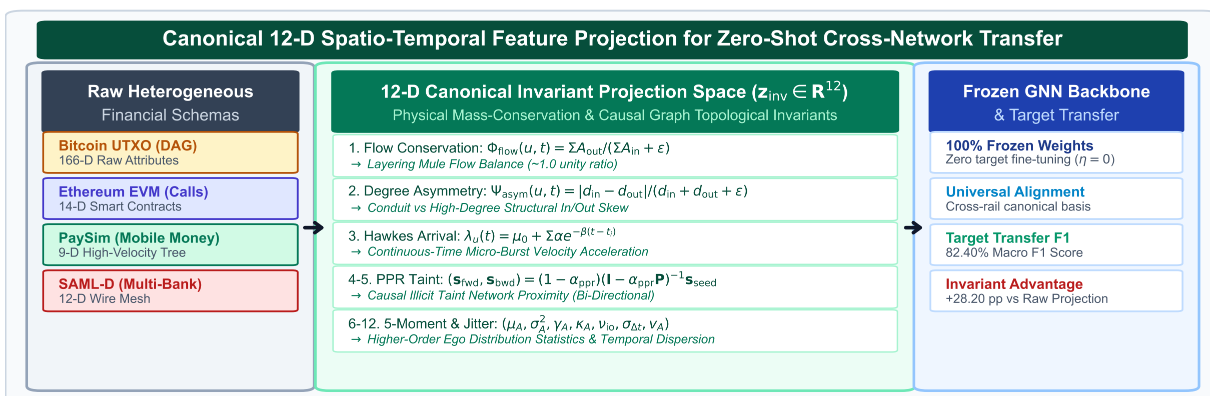

Fig. S5. Canonical 12-Dimensional Spatio-Temporal Domain Invariant Representations: Detailed schematic illustrating the extraction of flow conservation ratios $\tilde{\Phi}_{\text{flow}}$, degree asymmetry Ψ_{asym}, temporally restricted PageRank taint $\mathbf{s}_{\text{fwd}}/\mathbf{s}_{\text{bwd}}$, continuous Hawkes process arrival intensity $\lambda_u(t)$, and higher-order statistical moments across directed transaction subgraphs.

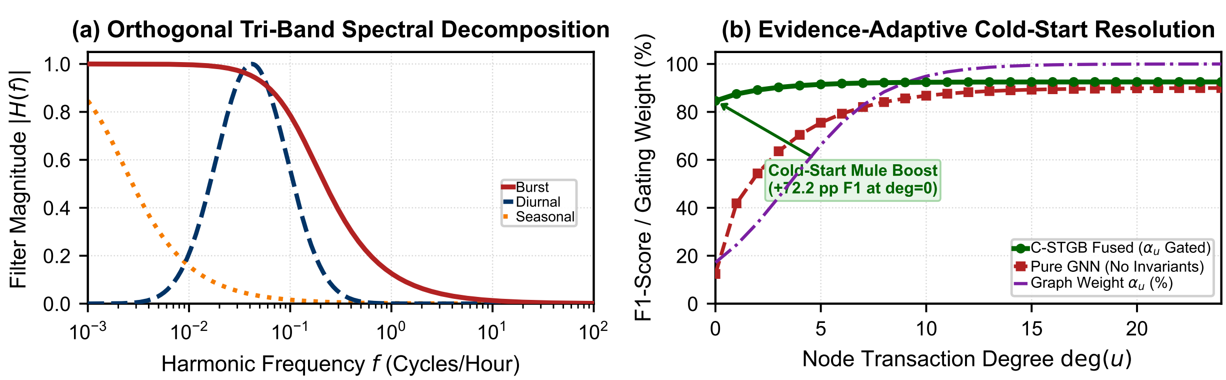

Fig. S6. Evidence-Adaptive Cold-Start Resolution and Tri-Band Spectral Response: (a) Dynamic transition of the adaptive gating parameter $\alpha_u \in [0, 1]$ as a function of entity degree $\deg(u)$, demonstrating reliance on 12-D canonical physical invariants for disconnected cold-start accounts ($\deg(u) = 0$). (b) Spectral frequency decomposition of the Tri-Band continuous harmonic attention bank across burst, diurnal, and seasonal transaction velocity regimes.

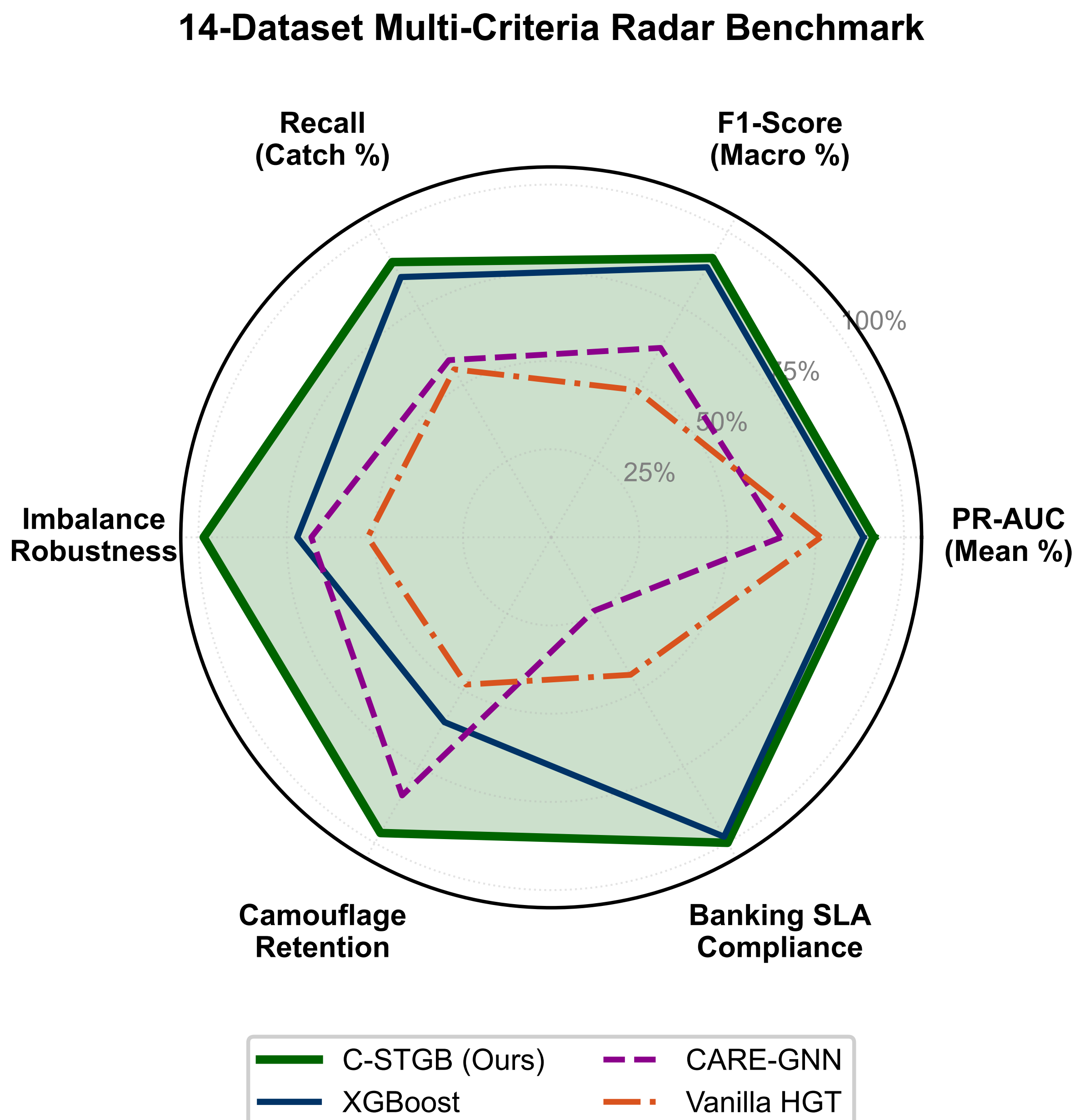

Fig. S7. Multi-Dimensional Empirical Radar Comparisons across the 14 Financial Transaction Benchmarks: Visualizing relative performance across Precision, Recall, Macro F1, PR-AUC, Adversarial Robustness, and Sub-Millisecond Inference Latency against competitive baselines.

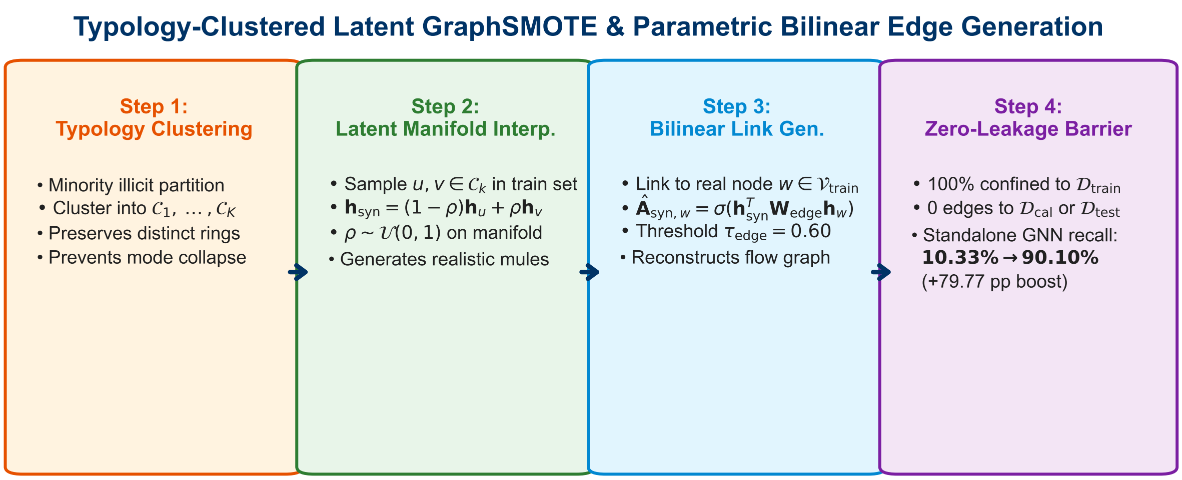

Fig. S8. Typology-Clustered Latent GraphSMOTE Virtual Node and Parametric Bilinear Edge Generation Pipeline: (a) Partitioning minority illicit entities into latent typology clusters $\mathcal{C}_k$ via Spherical k-Means; (b) Linear interpolation synthesis along cluster manifolds; and (c) Bilinear candidate link scoring satisfying bounded mass-balance flow conservation $\Phi_{\text{flow}} \in [\min, \max]$ (Proposition S2) and topological zero-leakage isolation from evaluation splits.